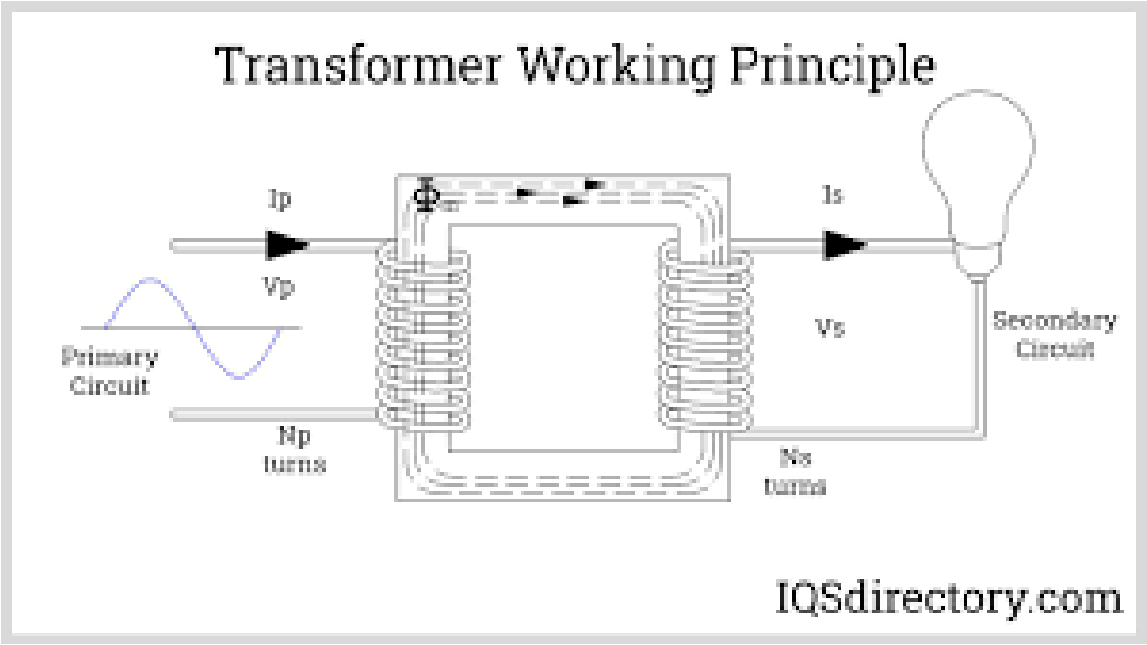
Exp 2

Transformer



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Code: 2300100 Sec: 5

Spring 25

Transformer Construction:

- Primary Winding: Connected to the input voltage source and creates a magnetic field in the core.
- Secondary Winding: Receives the induced voltage through mutual induction.
- Magnetic Core: Provides a path for magnetic flux and links the primary and secondary windings.
- Insulation: Separates the windings to avoid electrical contact and ensures safety.

Working Principle:

- Magnetic Field Generation: When AC voltage is applied to the primary winding, it creates a time-varying magnetic flux in the core.
- Mutual Induction: This changing magnetic flux links to the secondary winding and induces an EMF according to Faraday's Law of Electromagnetic Induction.
- Voltage Transformation: The induced voltage in the secondary depends on the turns ratio ($Vs/Vp = Ns/Np$).
- Load Connection: When a load is connected across the secondary winding, current flows, and power is transferred from primary to secondary.
- Efficiency Considerations: Transformer losses include copper loss (I^2R), core loss (hysteresis and eddy currents), and leakage flux.

Open Circuit Characteristics

- Circuit Setup: Connect the primary winding to the rated voltage supply with the secondary winding open. Use appropriate measuring instruments.
- Initial Operation: Energize the primary and allow the transformer to stabilize.
- Voltage & Current Measurement: Record the no-load primary current and the secondary voltage.
- Core Behavior: The no-load current is mainly used to magnetize the core, and the secondary voltage reflects the core's magnetic response.
- Reduced Voltage Testing: Repeat measurements with 80% of rated primary voltage and observe the secondary voltage behavior.

During Load Test explain what will happen to the primary current when load increases on the secondary side?

1. Increased Secondary Current: Draws more current through the load.
2. Magnetizing Effect: To maintain mutual flux, more current is drawn in the primary winding.
3. Voltage Regulation: Secondary voltage may drop due to internal impedance of the transformer.
4. Increased Copper Loss: Higher current causes more I^2R losses in both windings.

Lab readings

S.C Test	I= 34/5 A	p= 160 w	
S.C Test	V= 24 V	I= 42/5 A	p= 180 w
O.C Test	V= 99.4 V	I= 21/100 A	p= 6.5w

